

Derivation of the Geodesic Deviation Equation

Mriganka Raj Deka

1 Introduction and Physical Motivation

The geodesic deviation equation describes how nearby freely falling particles accelerate relative to one another due to the curvature of spacetime. We are interested in studying the relative acceleration between nearby geodesics. In flat spacetime, parallel geodesics remain parallel. However, in curved spacetime, nearby geodesics may converge or diverge, and this behavior is governed by the Riemann curvature tensor.

2 Setup and Definitions

Consider a congruence of nearby geodesics parametrized by two parameters:

- t : parameter along each geodesic
- s : parameter labeling nearby geodesics

The spacetime coordinates are given by

$$x^\mu = x^\mu(s, t)$$

We define two vector fields:

$$T^\mu = \frac{\partial x^\mu}{\partial t}$$

and

$$S^\mu = \frac{\partial x^\mu}{\partial s}$$

where:

- T is the rate at which a point p travels along the geodesic
- S is the rate at which a point p moves closer or further apart along the normal connecting two geodesics

Diagram of the Geodesic Congruence

3 Relative Velocity

The relative velocity is the rate at which the separation vector S changes along the geodesic's direction. This means we should parallel transport S along one of the geodesics and compare it

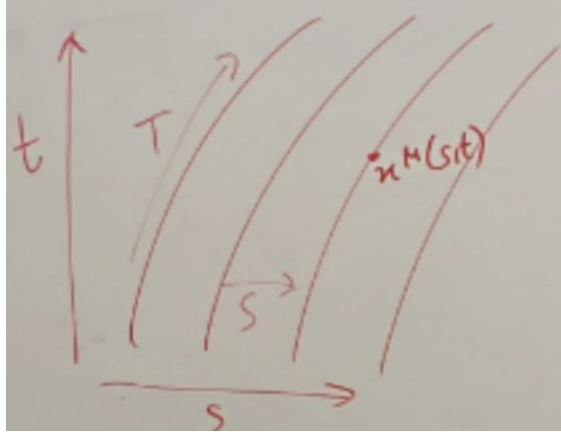


Figure 1: A congruence of nearby geodesics parametrized by t and s .

with the separation vector at that point. This means we must calculate the directional covariant derivative of S along T . Thus, the relative velocity is

$$V^\mu = \frac{D}{dt} S^\mu$$

Using the directional covariant derivative,

$$V^\mu = \frac{dx^\rho}{dt} \nabla_\rho S^\mu$$

which gives

$$V^\mu = T^\rho \nabla_\rho S^\mu \quad (i)$$

4 Relative Acceleration

The relative acceleration is defined as

$$A^\mu = \frac{D}{dt} V^\mu$$

Using equation (i),

$$A^\mu = T^\nu \nabla_\nu V^\mu = T^\nu \nabla_\nu (T^\rho \nabla_\rho S^\mu) \quad (ii)$$

5 Lie Bracket Condition

The commutator (Lie bracket) between the vector fields S and T vanishes:

$$[S, T] = 0$$

This is because of Clairaut's theorem which states that partial derivatives commute:

$$\frac{\partial}{\partial s} \frac{\partial}{\partial t} = \frac{\partial}{\partial t} \frac{\partial}{\partial s}$$

The μ -th component of the Lie bracket is

$$[S, T]^\mu = [(S^\rho \partial_\rho)(T^\mu \partial_\mu) - (T^\rho \partial_\rho)(S^\mu \partial_\mu)]$$

$$[S, T] = [(S^\rho \partial_\rho T^\mu) \partial_\mu + S^\rho T^\mu \partial_\rho \partial_\mu - (T^\rho \partial_\rho S^\mu) \partial_\mu - T^\rho S^\mu \partial_\rho \partial_\mu]$$

$$[S, T] = \{[(S^\rho \partial_\rho T^\mu) - (T^\rho \partial_\rho S^\mu)] \partial_\mu\}$$

$$[S, T]^\mu = S^\rho \partial_\rho T^\mu - T^\rho \partial_\rho S^\mu$$

Hence,

$$S^\rho \partial_\rho T^\mu = T^\rho \partial_\rho S^\mu$$

Generalizing to curved spacetime,

$$S^\rho \nabla_\rho T^\mu = T^\rho \nabla_\rho S^\mu \quad (\text{iii})$$

6 Expanding the Relative Acceleration

Substituting equation (iii) into equation (ii),

$$A^\mu = T^\nu \nabla_\nu (S^\rho \nabla_\rho T^\mu)$$

Using the product rule,

$$A^\mu = (T^\nu \nabla_\nu S^\rho) \nabla_\rho T^\mu + T^\nu S^\rho \nabla_\nu \nabla_\rho T^\mu$$

Using equation (iii) again,

$$T^\nu \nabla_\nu S^\rho = S^\nu \nabla_\nu T^\rho$$

Therefore,

$$A^\mu = (S^\nu \nabla_\nu T^\rho) \nabla_\rho T^\mu + T^\nu S^\rho \nabla_\nu \nabla_\rho T^\mu$$

Now use the commutator of covariant derivatives:

$$\nabla_\nu \nabla_\rho T^\mu = \nabla_\rho \nabla_\nu T^\mu + R^\mu_{\sigma\nu\rho} T^\sigma$$

Substituting,

$$A^\mu = (S^\nu \nabla_\nu T^\rho) \nabla_\rho T^\mu + T^\nu S^\rho (\nabla_\rho \nabla_\nu T^\mu + R^\mu_{\sigma\nu\rho} T^\sigma)$$

Rearranging,

$$A^\mu = S^\rho \nabla_\rho (T^\nu \nabla_\nu T^\mu) + R^\mu_{\sigma\nu\rho} T^\sigma T^\nu S^\rho \quad (\text{iv})$$

7 Using the Geodesic Equation

For a geodesic,

$$T^\nu \nabla_\nu T^\mu = 0$$

since freely falling observers have zero proper acceleration. Therefore, equation (iv) becomes

$$A^\mu = R^\mu_{\sigma\nu\rho} T^\sigma T^\nu S^\rho$$

Finally,

$$\boxed{\frac{D^2 S^\mu}{Dt^2} = R^\mu_{\sigma\nu\rho} T^\sigma T^\nu S^\rho}$$

This is the geodesic deviation equation.

8 Physical Interpretation

The geodesic deviation equation shows that spacetime curvature produces relative acceleration between nearby freely falling particles.

Thus:

- In flat spacetime, nearby geodesics remain parallel.
- In curved spacetime, geodesics may converge or diverge.
- The Riemann curvature tensor encodes tidal gravitational effects.

This equation hence provides us with an operational meaning to curvature: gravity manifests itself through differential free fall.